Projekt 2 HNS

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Dataset Pneumatiky

Trénovací dataset: 1324
Validačný dataset: 332





Štruktúra siete

```
def conv_block(in_ch, out_ch):
    return nn.Sequential(
        nn.Conv2d(in_ch, out_ch,
kernel_size=3, stride=1,
padding=1),
        nn.BatchNorm2d(out_ch),
        nn.ReLU(inplace=True),
        nn.Conv2d(out_ch,
out_ch, kernel_size=3,
stride=1, padding=1),
        nn.BatchNorm2d(out_ch),
        nn.ReLU(inplace=True)
    )

self.layer1 = conv_block(3, 64)
self.pool1 = nn.MaxPool2d(2)
self.dropout1 =
nn.Dropout2d(0.1)

self.layer2 = conv_block(64, 128)
self.pool2 = nn.MaxPool2d(2)
self.dropout2 = nn.Dropout2d(0.2)

self.layer3 = conv_block(128, 256)
self.pool3 = nn.MaxPool2d(2)
self.dropout3 = nn.Dropout2d(0.3)

self.layer4 = conv_block(256, 512)
self.dropout4 = nn.Dropout2d(0.4)

self.pool_avg = nn.AdaptiveAvgPool2d((1, 1))
self.pool_max = nn.AdaptiveMaxPool2d((1, 1))

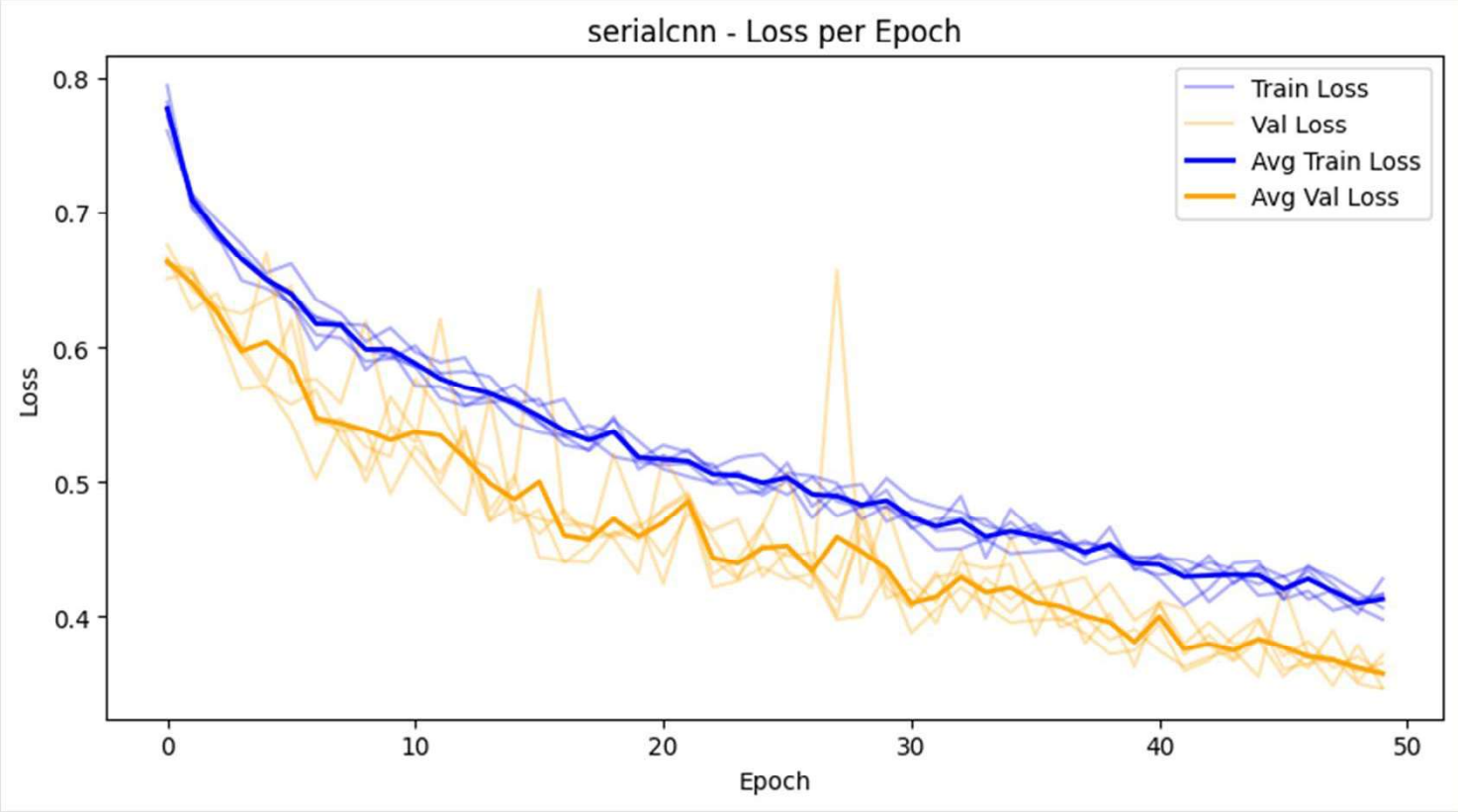
self.classifier = nn.Sequential(
    nn.Linear(512 * 2, 256),
    nn.ReLU(),
    nn.Dropout(0.5),
    nn.Linear(256, num_classes)
```

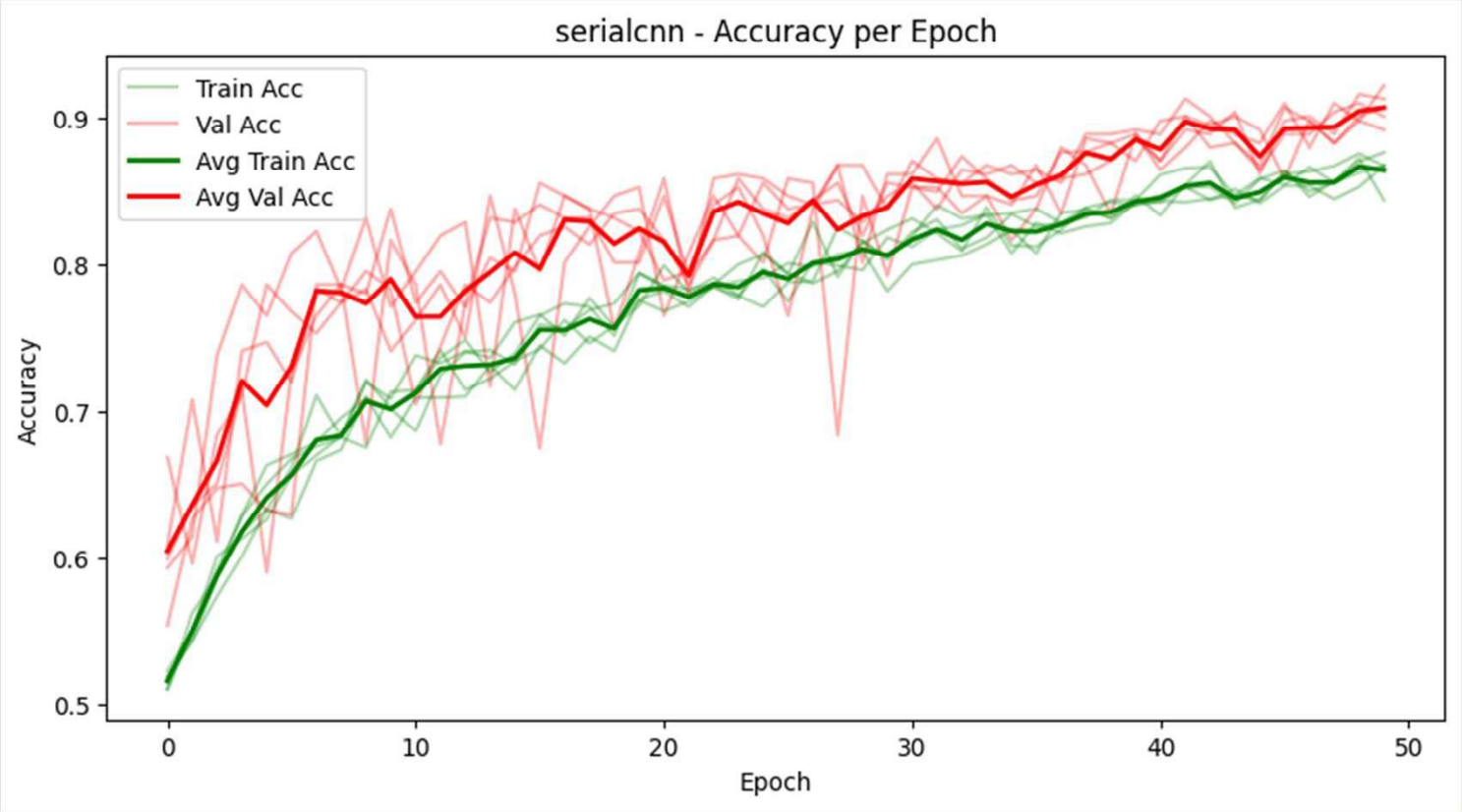
Parametre

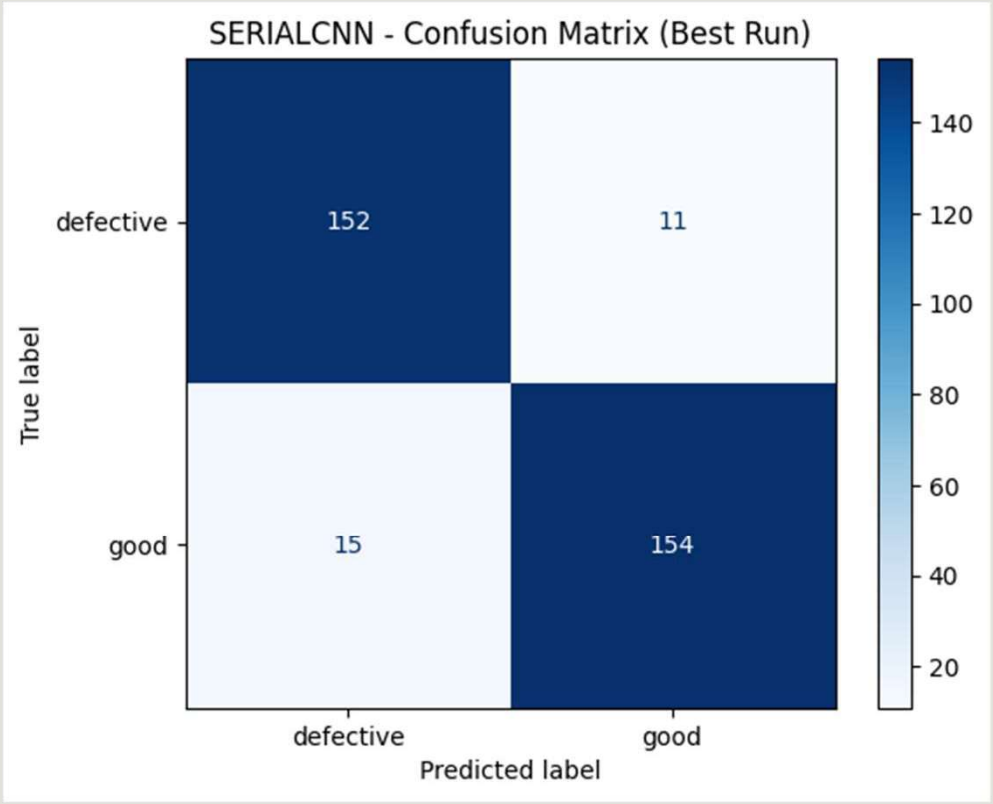
criterion = nn.CrossEntropyLoss(label_smoothing=0.1)

optimizer = optim.Adam(filter(lambda p: p.requires_grad, model.parameters()), lr=0.0001)

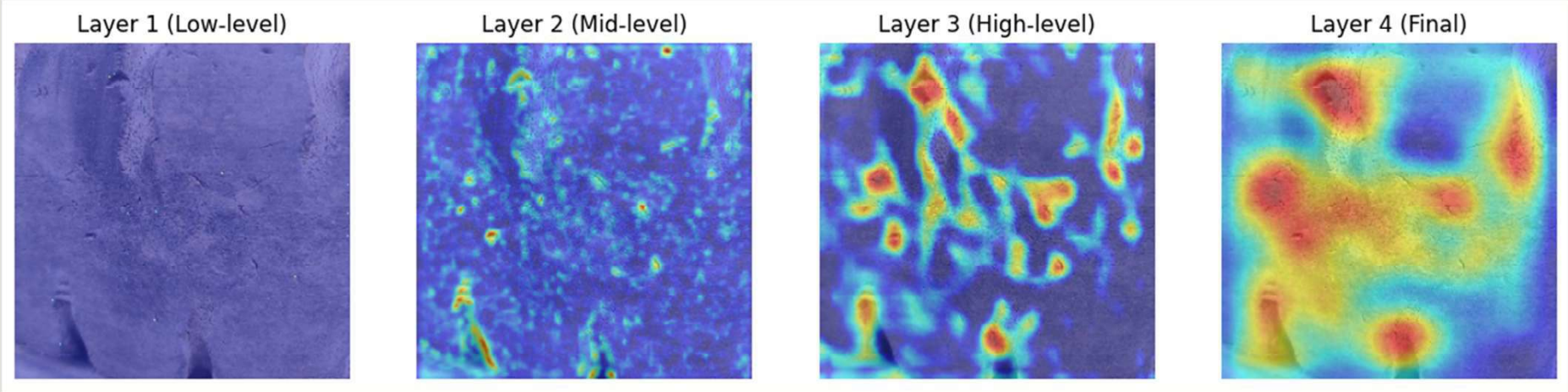
num_epoch = 50

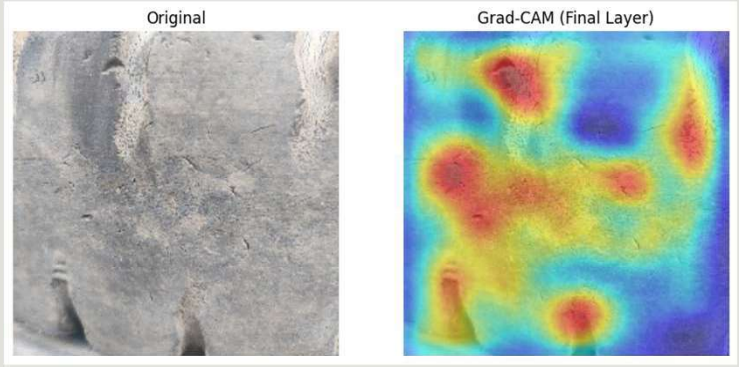

serialcnn - Loss per Epoch
Train Loss
Val Loss
Avg Train Loss
Avg Val Loss
Loss
Epoch
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50

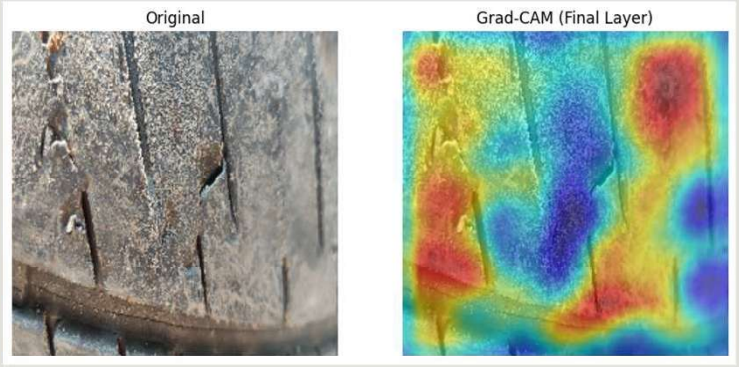

serialcnn - Accuracy per Epoch
Train Acc
Val Acc
Avg Train Acc
Avg Val Acc
Accuracy
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
Epoch

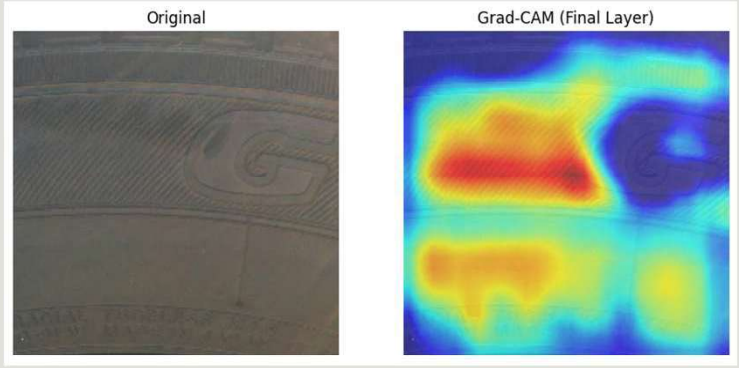

SERIALCNN - Confusion Matrix (Best Run)
defective
good
True label
152
11
15
154
defective
good
Predicted label
140
120
100
80
60
40
20

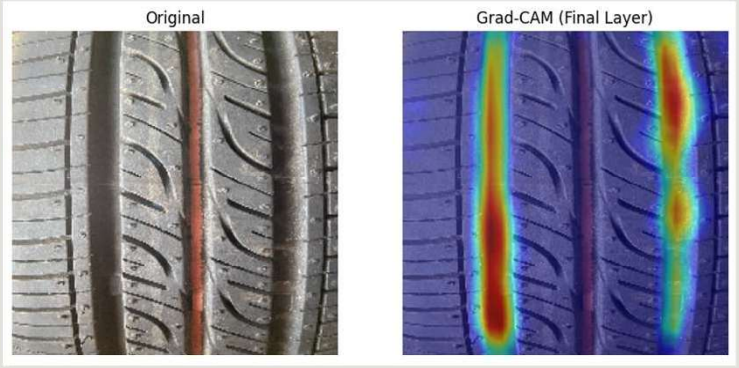
GradCAM

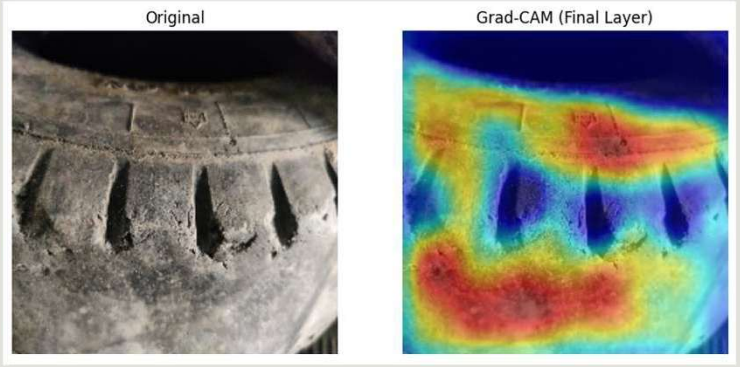

Layer 1 (Low-level)
Layer 2 (Mid-level)
Layer 3 (High-level)
Layer 4 (Final)


Original
Grad-CAM (Final Layer)


Original
Grad-CAM (Final Layer)


Original
Grad-CAM (Final Layer)


Original
Grad-CAM (Final Layer)


Original
Grad-CAM (Final Layer)